US 20250285141A1

(19) United States

(12) Patent Application Publication

KAWASE et al.

(10) Pub. No.: US 2025/0285141 A1

(43) Pub. Date: Sep. 11, 2025

(54) **INFORMATION PROCESSING DEVICE, CONTENT SWITCHING METHOD, VEHICLE, AND PROGRAM**

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(21) Appl. No.: **19/013,450**

(22) Filed: **Jan. 8, 2025**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (JP) 2024-036169

Publication Classification

(51) **Int. Cl.**
G06Q 30/0251 (2023.01)

(52) **U.S. Cl.**
CPC *G06Q 30/0266* (2013.01)

(57) **ABSTRACT**

An information processing device suspends provision of non-advertisement content and starts provision of advertisement content, in a case where (a) a first determining section determines that a traveling body has stopped traveling while the non-advertisement content is being provided and (b) a second determining section determines to permit suspension of provision of the non-advertisement content.

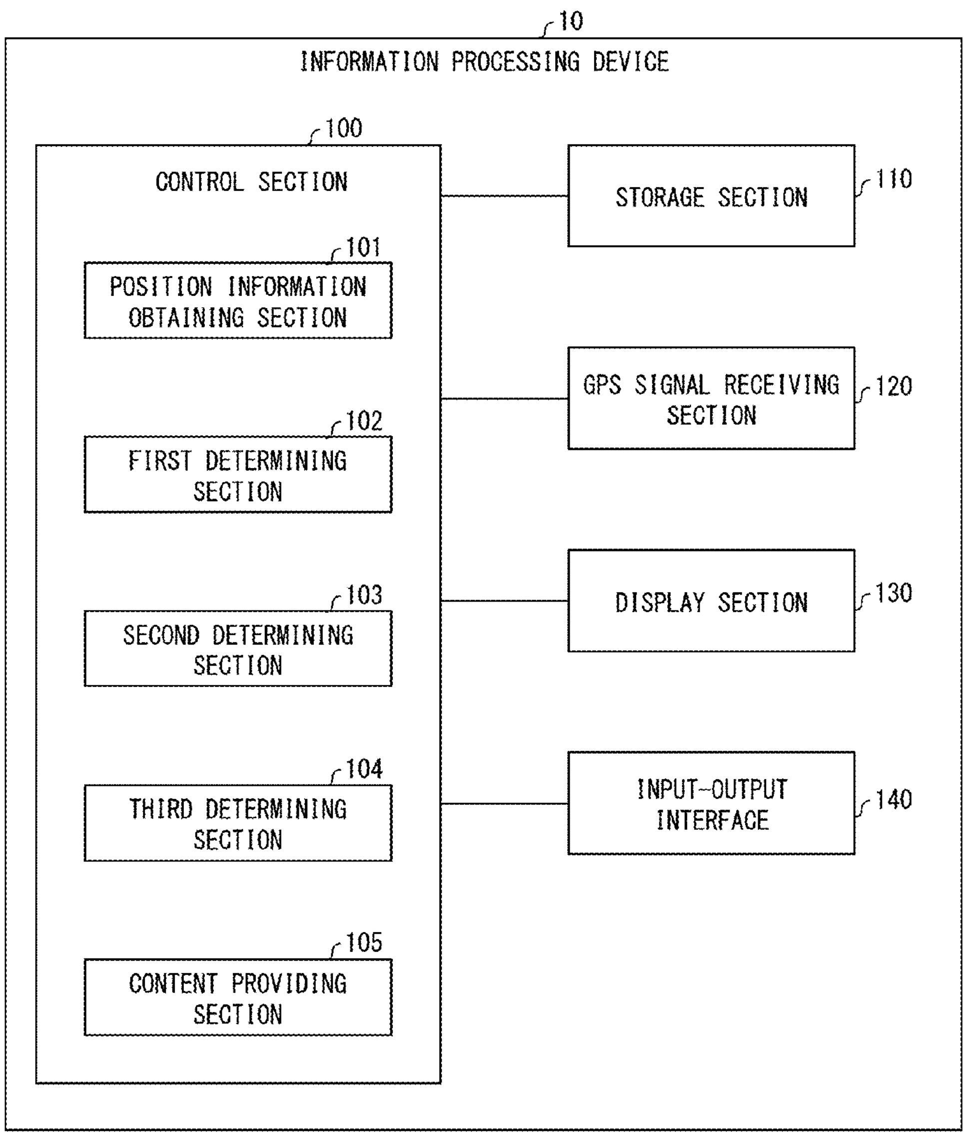


FIG. 1

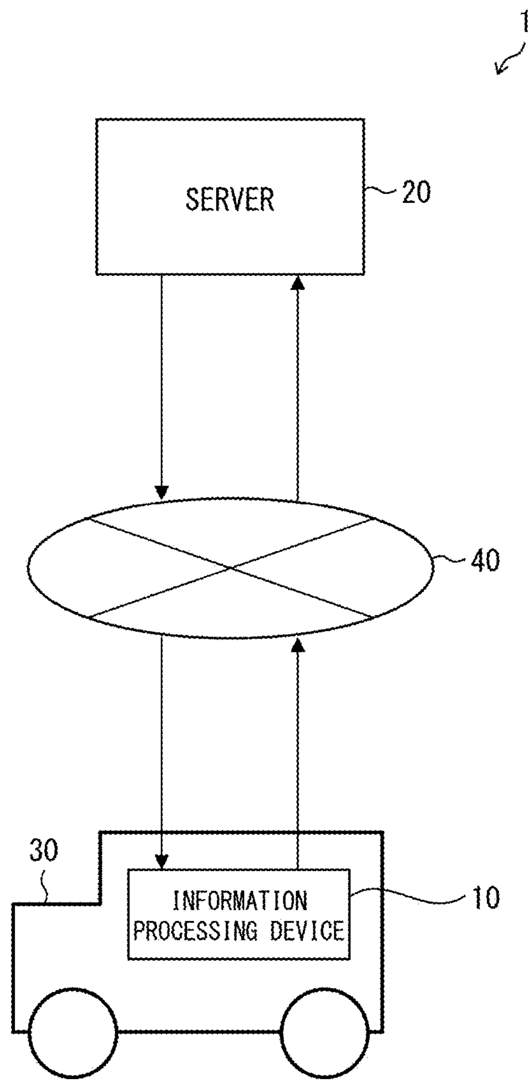


FIG. 2

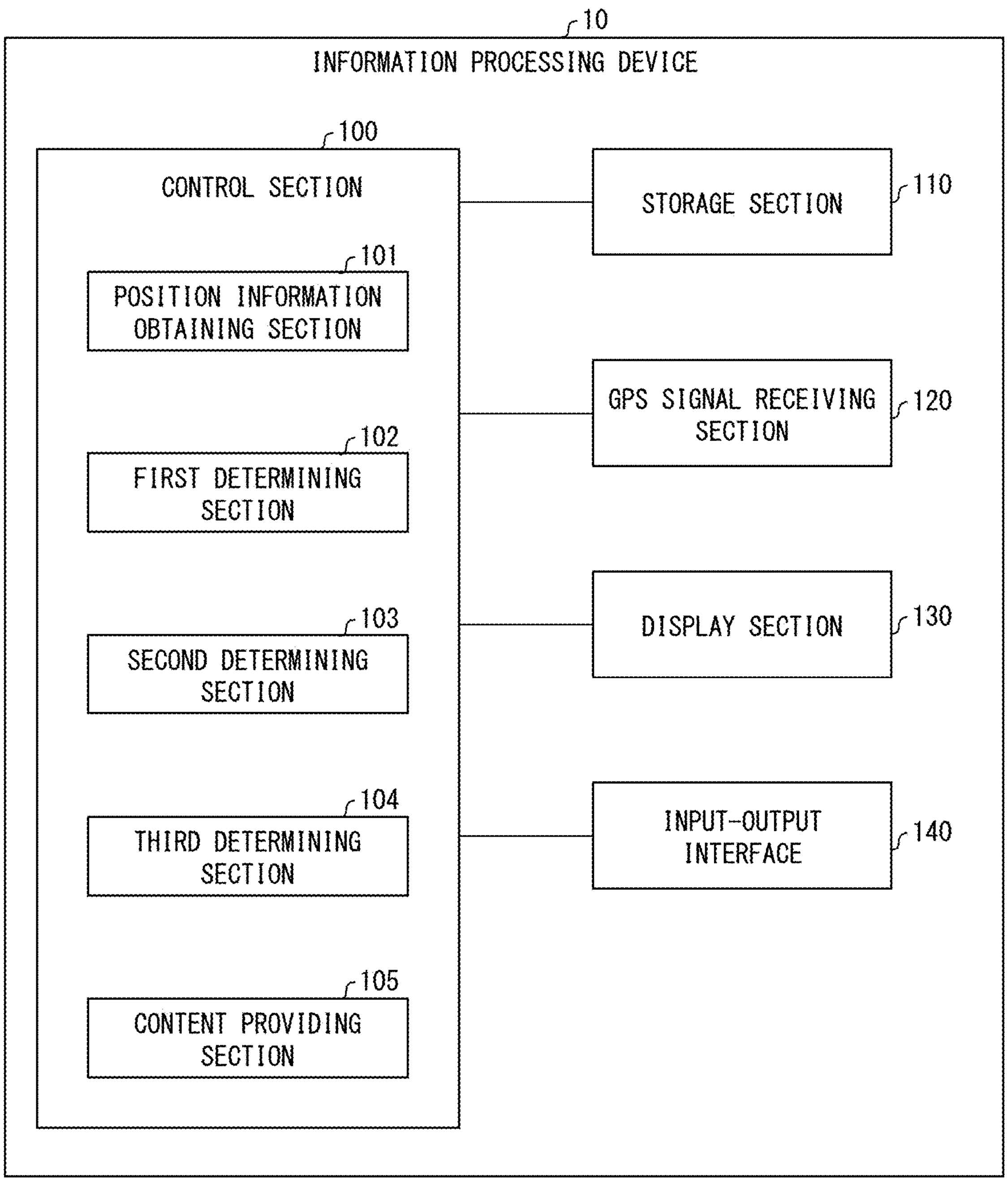

10
INFORMATION PROCESSING DEVICE
100
CONTROL SECTION
101
POSITION INFORMATION OBTAINING SECTION
102
FIRST DETERMINING SECTION
103
SECOND DETERMINING SECTION
104
THIRD DETERMINING SECTION
105
CONTENT PROVIDING SECTION
STORAGE SECTION
110
GPS SIGNAL RECEIVING SECTION
120
DISPLAY SECTION
130
INPUT-OUTPUT INTERFACE
140

FIG. 3

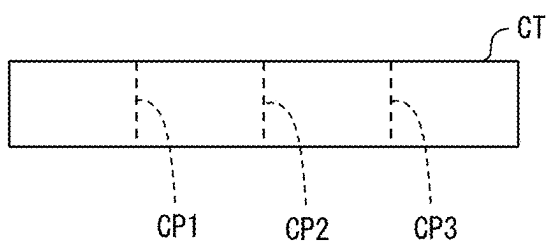
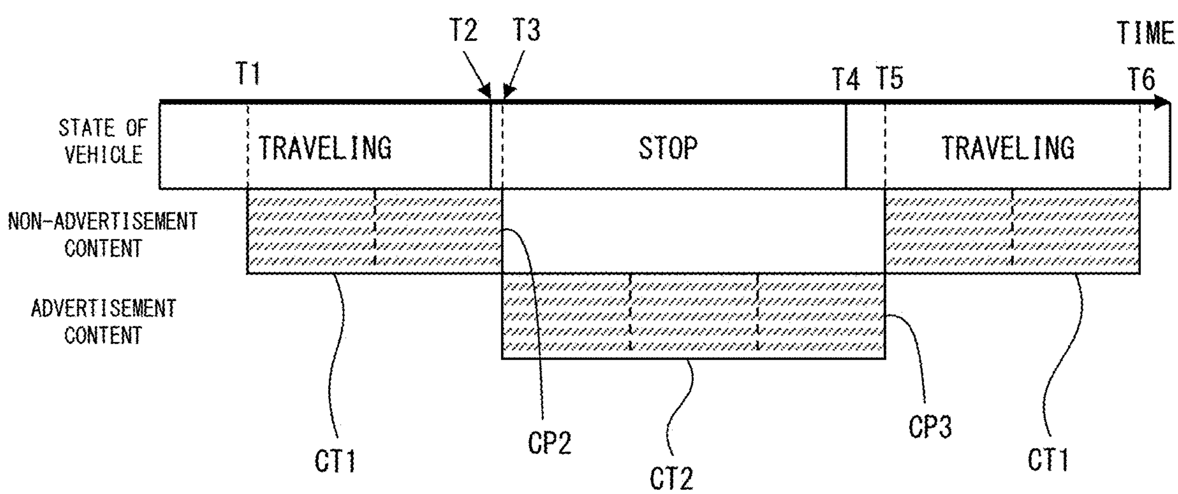

CT
CP1
CP2
CP3

FIG. 4


T2
T3
TIME
T1
T4
T5
T6
STATE OF VEHICLE
TRAVELING
STOP
TRAVELING
NON-ADVERTISEMENT CONTENT
ADVERTISEMENT CONTENT
CT1
CP2
CT2
CP3
CT1

INFORMATION PROCESSING DEVICE, CONTENT SWITCHING METHOD, VEHICLE, AND PROGRAM

[0001] This Nonprovisional application claims priority under 35 U.S.C. § 119 on Patent Application No. 2024-036169 filed in Japan on Mar. 8, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an information processing device, a content switching method, a vehicle, and a program.

BACKGROUND ART

[0003] Patent Literature 1 discloses a content providing method according to which, when it is determined that a vehicle stops, an estimated stopping period is calculated and advertisement content which can be played back within the estimated stopping period is output.

CITATION LIST

Patent Literature

[Patent Literature 1]

[0004] Japanese Patent Application Publication, Tokukai, No. 2021-039503

SUMMARY OF INVENTION

Technical Problem

[0005] According to the content providing method disclosed in Patent Literature 1, it is necessary to calculate the estimated stopping time period of the vehicle. An aspect of the present has an object to provide advertisement content while a traveling body stops, even without estimating a time period during which the traveling body stops.

Solution to Problem

[0006] In order to attain the above object, an information processing device in accordance with an aspect of the present disclosure is an information processing device that provides content including advertisement content and non-advertisement content to a passenger of a traveling body, the information processing device including: a first determining section configured to determine whether the traveling body is traveling or the traveling body has stopped traveling; and a second determining section configured to determine, on a basis of whether a first time period elapsed from a timing of starting provision of the non-advertisement content is a given first suspension time period, whether to permit suspension of provision of the non-advertisement content, provision of the non-advertisement content being suspended and provision of the advertisement content being started, in a case where (a) the first determining section determines that the traveling body has stopped traveling while the non-advertisement content is being provided and (b) the second determining section determines to permit suspension of provision of the non-advertisement content.

[0007] In order to attain the above object, a content switching method in accordance with another aspect of the present disclosure is a content switching method that switches content which is to be provided to a passenger of a traveling body between advertisement content and non-advertisement content, the content switching method including: suspending provision of the non-advertisement content and starting provision of the advertisement content, in a case where (a) the traveling body stops while the non-advertisement content is being provided and (b) a time period elapsed from a timing of starting provision of the non-advertisement content is a given time period.

[0008] An information processing device in accordance with each aspect of the present disclosure may be realized by a computer. In this case, the present disclosure encompasses: a program for the information processing device, the program causing a computer to operate as the sections (software elements) included in the information processing device SO that the information processing device is realized by the computer; and a computer-readable storage medium in which the program is stored.

Advantageous Effects of Invention

[0009] In accordance with an aspect of the present disclosure, it is possible to provide advertisement content while a traveling body stops, even without estimating a time in which the traveling body stops.

BRIEF DESCRIPTION OF DRAWINGS

[0010] FIG. **1** is a block diagram illustrating an example of a configuration of a content providing system that uses an information processing device in accordance with an embodiment of the present disclosure.

[0011] FIG. **2** is a view illustrating an example of a configuration of an information processing device in accordance with an embodiment of the present disclosure.

[0012] FIG. **3** is a view used to explain a second determining section and a third determining section.

[0013] FIG. **4** is a view used to explain a content switching method in accordance with an embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

[0014] FIG. **1** is a block diagram illustrating an example of a configuration of a content providing system that uses an information processing device in accordance with an embodiment of the present disclosure. A content providing system **1** shown in FIG. **1** includes an information processing device **10** and a server **20**. For example, the information processing device **10** is disposed inside a vehicle **30**, and provides content to a passenger of the vehicle **30**. The vehicle **30** is one example of the traveling body, and is, for example, an automobile, a bus, or a taxi. The content is the one that can be sensed by the passenger with five senses. The content includes a video, an audio, a vibration, an odor, and/or mist, for example.

[0015] The server **20** is disposed outside the information processing device **10**. The server **20** has a function to transmit, to the information processing device **10**, data of the content that the information processing device **10** provides to the passenger. The data of the content includes: guiding content for providing information about (i) a facility existing in an area surrounding a road on which the vehicle **30** travels, (ii) a building that is visually recognizable through a window of the vehicle **30**, and/or (iii) a landmark(s) such as a river and/or a mountain; and advertising content for advertisement. The guiding content is one example of the non-advertisement content.

[0016] The data relating to the guiding content includes (i) information indicative of a position (a latitude, a longitude, and/or an altitude) of a subject to be guided (i.e., a guiding subject) such as a facility and/or a landmark, (ii) data indicative of a video and/or an audio relating to the guiding subject, and (iii) information about a playback time period of a video and/or an audio.

[0017] The information processing device **10** and the server **20** are connected to a network **40** such as a wireless communication network, an Internet network, or a telephone line network. For convenience of explanation, FIG. **1** illustrates a single information processing device **10**. Alternatively, the content providing system **1** may include a plurality of information processing devices **10**. Further, the information processing device **10** and the server **20** may be connected to, via the network **40** and/or the like, an information terminal, a server, and/or the like (not illustrated). For example, the information processing device **10** and the server **20** may be connected to an information terminal possessed by the passenger of the vehicle **30**. The information processing device **10** obtains data of non-advertisement content such as the guiding content and advertisement content from the server **20** via the network **40**.

[0018] FIG. **2** is a view illustrating an example of a configuration of the information processing device in accordance with the embodiment of the present disclosure. As shown in FIG. **2**, the information processing device **10** includes a control section **100**, a storage section **110**, a GPS signal receiving section **120**, a display section **130**, and an input-output interface **140**.

[0019] The control section **100** includes, e.g., a central processing unit (CPU) and a main storage device. The storage section **110** includes, e.g., a hard disk drive (HDD) and/or a solid state drive (SSD). In the storage section **110**, a program to be executed by the control section **100** and data of content transmitted from the server **20** are stored.

[0020] The GPS signal receiving section **120** receives a signal from a GPS satellite. The display section **130** is one example of the output device, and can be used to provide content such as a still image and/or a moving image. The input-output interface **140** is, for example, a universal serial bus (USB) terminal and/or a local area network (LAN) terminal. The information processing device **10** is connected, via the input-output interface **140**, to a vehicle communication network such as a controller area network (CAN). The information processing device **10** is connectable to the network **40** via CAN.

[0021] By executing the program stored in the storage section **110**, the control section **100** functions as a position information obtaining section **101**, a first determining section **102**, a second determining section **103**, a third determining section **104**, and a content providing section **105**.

[0022] The position information obtaining section **101** obtains position information indicative of a current position of the vehicle **30**. For example, the position information obtaining section **101** obtains, on the basis of a signal that the GPS signal receiving section **120** has received from a GPS satellite, position information indicative of a current position of the vehicle **30**. The position information obtaining may be connected to CAN via the input-output interface **140** and may be configured to obtain vehicle information about a vehicle speed of the vehicle **30**, an acceleration of the vehicle **30**, and amounts of operation of an accelerator pedal, a brake pedal, and a steering wheel of the vehicle **30** and to use the vehicle information to calculate the current position of the vehicle **30**.

[0023] The first determining section **102** determines whether the vehicle **30** is traveling or the vehicle **30** has stopped traveling. Examples of the state in which "the vehicle **30** has stopped traveling" include a state in which the vehicle **30** stops and a state in which the vehicle **30** is parked. For example, in a case where the position information obtained by the position information obtaining section **101** does not change, the first determining section **102** determines that the vehicle **30** has stopped traveling. Meanwhile, in a case where the position information obtained by the position information obtaining section **101** changes, the first determining section **102** determines that the vehicle **30** is traveling.

[0024] The content providing section **105** provides content through, e.g., the display section **130** of the information processing device **10**, an output device provided to the vehicle **30**, or an output device possessed by the passenger.

[0025] FIG. **3** is a view used to explain the second determining section **103** and the third determining section **104**. The data of the content transmitted from the server **20** to the information processing device **10** includes at least one suspension time period during which suspension of provision of content is permitted. Content CT shown in FIG. **3** includes three suspension time periods CP**1**, CP**2**, and CP**3**.

[0026] In a case where the content CT is non-advertisement content such as guiding content, each of the suspension time period periods CP**1**, CP**2**, and CP**3** is one example of the first suspension time period. The second determining section **103** shown in FIG. **2** determines, on the basis of whether or not a time period elapsed from a timing of starting provision of non-advertisement content is any of the suspension time periods CP**1**, CP**2**, and CP**3**, whether to permit suspension of provision of the non-advertisement content. In a case where the period of time elapsed from the timing of starting provision of the non-advertisement content is any of the suspension time periods CP**1**, CP**2**, and CP**3** for the non-advertisement content, the second determining section **103** determines to permit suspension of provision of the non-advertisement content. Meanwhile, in a case where the period of time elapsed from the timing of starting provision of the non-advertisement content is not the suspension time period CP**1**, CP**2**, or CP**3** for the non-advertisement content, the second determining section **103** determines not to permit suspension of provision of the non-advertisement content.

[0027] In a case where the content CT is advertisement content, the suspension time periods CP**1**, CP**2**, and CP**3** are examples of a second suspension time period. The third determining section **104** shown in FIG. **2** determines, on the basis of whether or not a time period elapsed from a timing of starting provision of the advertisement content is any of the suspension time periods CP**1**, CP**2**, and CP**3**, whether to permit suspension of provision of the advertisement content. In a case where the time period elapsed from the timing of starting provision of the advertisement content is any of the suspension time periods CP**1**, CP**2**, and CP**3** for the advertisement content, the third determining section **104** determines to permit suspension of provision of the advertisement content. Meanwhile, in a case where the time period elapsed from the timing of starting provision of the advertisement content is not the suspension time period CP**1**, CP**2**, or CP**3** for the advertisement content, the third determining section **104** determines not to permit suspension of provision of the advertisement content.

[0028] Each of the suspension time periods CP**1**, CP**2**, and CP**3** is provided at a timing which provides a good separation in the content CT. Providing the "good separation in the content CT" means that suspending provision of the content CT with a low possibility of giving the passenger an uncomfortable feeling and/or a strange feeling. For example, in a case where the content CT is video content, the video content may be suspended at a timing when a video scene is switched from one to another. This can decrease the possibility of giving the passenger an uncomfortable feeling and/or a strange feeling. In a case where the content CT is audio content, the audio content may be suspended on the basis of, e.g., a breath group in the audio which is being output and/or a section in music. This can decrease the possibility of giving the passenger an uncomfortable feeling and/or a strange feeling.

[0029] In a case where the content CT is prepared so that a good separation is given at each given time period (e.g., 15 seconds), the suspension time periods CP**1**, CP**2**, and CP**3** may be set in advance in the process of preparing the content CT.

[0030] Alternatively, the suspension time periods CP**1**, CP**2**, and CP**3** may be set on the basis of a similarity between pieces of content which are to be output consecutively during or before provision of the content CT. For example, in a case where the content CT is video content, feature vectors of frame images in respective frames may be calculated, and frames in each of which a cosine similarity between two consecutive frames is not more than a given threshold may be regarded as the suspension time period CP**1**, CP**2**, or CP**3**.

[0031] FIG. **4** is a view used to explain a content switching method in accordance with an embodiment of the present disclosure. FIG. **4** illustrates an example of changes over time in (i) a state of the vehicle **30**, (ii) a state of provision of the non-advertisement content CT**1**, and (iii) a state of provision of the advertisement content CT**2**.

[0032] In FIG. **4**, the vehicle **30** stops at a timing T**2**, and starts traveling at a timing T**4**. Thus, the first determining section **102** determines that the vehicle **30** stops and is not traveling during a period from the timing T**2** to the timing T**4**, and determines that the vehicle **30** is traveling before the timing T**2** and after the timing T**4**.

[0033] The information processing device **10** starts providing the non-advertisement content CT**1** to the passenger of the vehicle **30** at a timing T**1**, which is before the timing T**2**. When the vehicle stops at the timing T**2**, a first time period elapsed from the timing T**1** of starting provision of the non-advertisement content CT**1** is not the suspension time period CP**1**, CP**2**, or CP**3** for the non-advertisement content CT**1**. Thus, at the timing T**2**, the second determining section **103** determines not to permit suspension of provision of the non-advertisement content CT**1**. At a timing T**3** between the timings T**2** and T**4**, the first time period reaches the suspension time period CP**2** for the non-advertisement content CT**1**; therefore, the second determining section **103** determines to permit suspension of provision of the non-advertisement content CT**1**. At the timing T**3**, the information processing device **10** suspends provision of the non-advertisement content CT**1**, and starts provision of the advertisement content CT**2**.

[0034] By suspending provision of the non-advertisement content CT**1** after the suspension time period for the non-advertisement content CT**1** comes rather than by suspending provision of the non-advertisement content CT**1** immediately at the time when the vehicle **30** stops, it is possible to prevent an uncomfortable feeling and/or a strange feeling from being given to the passenger of the vehicle **30**. By providing the advertisement content CT**2** mainly while the vehicle **30** stops, it is possible to avoid a situation in which the passenger cannot enjoy a scenery through the window of the vehicle. Further, by providing the advertisement content CT**2** while the vehicle **30** stops, it is possible to increase the change that the advertisement content CT**2** catches the passenger's attention.

[0035] When the vehicle starts traveling at the timing T**4**, a second time period elapsed from the timing T**3** of starting provision of the advertisement content CT**2** is not the suspension time period CP**1**, CP**2**, or CP**3** for the advertisement content CT**2**. Thus, the third determining section **104** determines not to permit suspension of provision of the advertisement content CT**2** at the timing T**4**. At a timing T**5**, which is later than the timing T**4**, the second time period reaches the suspension time period CP**3** for the advertisement content CT**2**; therefore, the third determining section **104** determines to permit suspension of provision of the advertisement content CT**2**. At the timing T**5**, the information processing device **10** suspends provision of the advertisement content CT**2** and restarts provision of the non-advertisement content CT**1**. As shown in FIG. **4**, the information processing device **10** may start provision of the non-advertisement content CT**1** at the timing T**3** when provision of the non-advertisement content CT**1** is suspended. In FIG. **4**, provision of the non-advertisement content CT**1** is completed at a timing T**6**, which is later than the timing T**5**.

Effects

[0036] As explained above, according to the information processing device **10** in accordance with the present embodiment, it is possible to attain the following effects.

[0037] The information processing device **10** provides the content including the advertisement content CT**2** and the non-advertisement content CT**1** to the passenger of the vehicle **30**, and includes the first determining section **102** and the second determining section **103**. The first determining section **102** determines whether the vehicle **30** is traveling or the vehicle **30** has stopped traveling. The second determining section **103** determines, on the basis of whether the first time period elapsed from the timing T**1** of starting provision of the non-advertisement content CT**1** is any of the suspension time periods CP**1**, CP**2**, and CP**3** for the non-advertisement content CT**1**, whether to permit suspension of provision of the non-advertisement content CT**1**. In a case where (a) the first determining section **102** determines that the vehicle **30** has stopped traveling while the non-advertisement content CT**1** is being provided and (b) the second determining section **103** determines to permit suspension of provision of the non-advertisement content CT**1** (T**3**), provision of the non-advertisement content CT**1** is suspended and provision of the advertisement content CT**2** is started. In accordance with the above embodiment, it is possible to provide the advertisement content CT2 when the vehicle 30 stops, even without estimating a time period during which the vehicle 30 stops.

[0038] The information processing device 10 further includes the third determining section 104. The third determining section 104 determines, on the basis of whether or not a second time period from the timing T3 of starting provision of the advertisement content CT2 is any of the suspension time periods CP1, CP2, and CP3 for the advertisement content CT2, whether to permit suspension of provision of the advertisement content CT2. In a case where (a) the first determining section 102 determines that the vehicle 30 is traveling while provision of the non-advertisement content CT1 is suspended and the advertisement content CT2 is being provided and (b) the third determining section 104 determines to permit suspension of provision of the advertisement content CT2, provision of the advertisement content CT2 is suspended. In accordance with the above embodiment, it is possible to suspend provision of the advertisement content CT2 when the vehicle 30 travels, even without giving a strange feeling to the passenger.

[0039] The information processing device 10 is configured as follows. In a case where (a) the first determining section 102 determines that the vehicle 30 is traveling while provision of the non-advertisement content CT1 is suspended and the advertisement content CT2 is being provided and (b) the third determining section 104 determines to permit suspension of provision of the advertisement content CT2, provision of the non-advertisement content CT1 whose provision has been suspended is restarted. In accordance with the above embodiment, provision of the non-advertisement content CT1 whose provision has been suspended is restated. This makes it easy to complete provision the non-advertisement content CT1.

[0040] The content switching method explained with reference to FIG. 4 is a method for switching content which is to be provided to the passenger of the vehicle 30 between the advertisement content CT2 and the non-advertisement content CT1. In a case where (a) the vehicle 30 stops while the non-advertisement content CT1 is being provided and (b) a time period elapsed from a timing of starting provision of the non-advertisement content CT1 is any of the given suspension time periods CP1, CP2, and CP3, provision of the advertisement content CT1 is suspended and provision of the advertisement content CT2 is started. According to the above embodiment, it is possible to provide the advertisement content CT2 when the vehicle 30 stops, even without estimating a time period during which the vehicle 30 stops.

[0041] The vehicle 30 includes the information processing device 10.

Software Implementation Example

[0042] The functions of the information processing device 10 (hereinafter, referred to as a "device") can be realized by a program for causing a computer to function as the device, the program causing the computer to function as the control blocks (particularly, the sections included in the control section 100) of the device.

[0043] In this case, the device includes, as hardware for executing the program, at least one control device (e.g., a processor) and at least one storage device (e.g., a memory). By causing the control device and the storage device to execute the program, the functions explained in the foregoing embodiments are realized.

[0044] The program may be recorded in one or more non-transitory computer-readable recording media. The recording media may or may not be included in the device. In the latter case, the program may be supplied to or made available to the device via any wired or wireless transmission medium.

[0045] Furthermore, some or all of the functions of the control blocks can also be realized by a logic circuit. For example, the present disclosure encompasses, in its scope, an integrated circuit in which a logic circuit that functions as each of the above-described control blocks is formed. In addition, the function of each of the control blocks can be realized by, for example, a quantum computer.

[0046] The processes described in the foregoing embodiments may be carried out by artificial intelligence (AI). In this case, AI may be operated in the control device, or may be operated in another device (e.g., an edge computer or a cloud server).

Variations

[0047] The foregoing embodiment has dealt with the case where the information processing device 10 is provided inside the vehicle 30. However, even in a case where an information processing device in accordance with an embodiment of the present disclosure is disposed inside a traveling body other than the vehicle 30, for example, a railroad train, an aircraft, or a ship, it is possible to provide similar effects.

[0048] In the foregoing embodiment, the information processing device 10 is configured such that data of content which is to be provided to the passenger is transmitted from the server 20 to the information processing device 10. Alternatively, however, the data of the content may be stored in the storage section 110 of the information processing device 10 in advance.

[0049] In the foregoing embodiment, the control section 100 of the information processing device 10 disposed inside the vehicle 30 functions as the first determining section 102, the second determining section 103, and the third determining section 104. However, the information processing device including the first determining section 102, the second determining section 103, and the third determining section 104 may be disposed outside the vehicle 30. For example, the server 20 may function as the information processing device. In a case where the server 20 is caused to function as the first determining section 102, the second determining section 103, and the third determining section 104, (i) position information of the vehicle 30 and/or (ii) vehicle information about a vehicle speed, an acceleration, and/or an amount of operation of an accelerator pedal of the vehicle 30 may be transmitted from the information processing device 10 to the server 20.

SUPPLEMENTARY REMARKS

[0050] The present disclosure is not limited to the embodiments above, but can be altered by a skilled person in the art within the scope of the claims. The present disclosure also encompasses, in its technical scope, any embodiment derived by combining technical means disclosed in differing embodiments as appropriate.

REFERENCE SIGNS LIST

[0051] **10**: information processing device

[0052] **20**: server

[0053] **30**: vehicle

[0054] **100**: control section

[0055] **102**: first determining section

[0056] **103**: second determining section

[0057] **104**: third determining section

[0058] CP**1**, CP**2**, CP**3**: suspension time period (first suspension time period, second suspension time period)

[0059] CT: content

[0060] CT**1**: non-advertisement content

[0061] CT**2**: advertisement content

[0062] T**1**, T**2**, T**3**, T**4**, T**5**, T**6**: timing

1. An information processing device that provides content including advertisement content and non-advertisement content to a passenger of a traveling body, the information processing device comprising:

a first determining section configured to determine whether the traveling body is traveling or the traveling body has stopped traveling; and

a second determining section configured to determine, on a basis of whether a first time period elapsed from a timing of starting provision of the non-advertisement content is a given first suspension time period, whether to permit suspension of provision of the non-advertisement content,

provision of the non-advertisement content being suspended and provision of the advertisement content being started, in a case where (a) the first determining section determines that the traveling body has stopped traveling while the non-advertisement content is being provided and (b) the second determining section determines to permit suspension of provision of the non-advertisement content.

2. The information processing device according to claim **1**, further comprising:

a third determining section configured to determine, on a basis of whether or not a second time period elapsed from a timing of starting provision of the advertisement content is a given second suspension time period, whether to permit suspension of provision of the advertisement content, wherein:

provision of the advertisement content is suspended, in a case where (a) the first determining section determines that the traveling body is traveling while provision of the non-advertisement content is suspended and the advertisement content is being provided and (b) the third determining section determines to permit suspension of provision of the advertisement content.

3. The information processing device according to claim **2**, wherein:

provision of the non-advertisement content whose provision has been suspended is restarted, in a case where (a) the first determining section determines that the traveling body is traveling while provision of the non-advertisement content is suspended and the advertisement content is being provided and (b) the third determining section determines to permit suspension of provision of the advertisement content.

4. A content switching method that switches content which is to be provided to a passenger of a traveling body between advertisement content and non-advertisement content, the content switching method comprising:

suspending provision of the non-advertisement content and starting provision of the advertisement content, in a case where (a) the traveling body stops while the non-advertisement content is being provided and (b) a time period elapsed from a timing of starting provision of the non-advertisement content is a given time period.

5. A vehicle comprising an information processing device recited in claim **1**.

6. A program for causing a computer to function as an information processing device recited in claim **1**, the program causing the computer to function as the first determining section and the second determining section.

* * * * *